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Methodology For The Demonstration Of Electrical Compatibility With DC (AC Immune) Track Circuits

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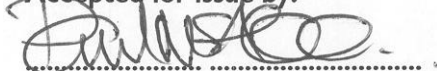
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IMPLEMENTATION

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I INTRODUCTION

I.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with a.c. immune d.c. track circuits on the a.c. railway on Network Rail Controlled Infrastructure. This is achieved by:

- providing a set of standard d.c. interference current limits for steady state interference;
- providing a set of d.c. interference voltage limits for steady state axle to axle interference;
- providing a methodology to determine whether low frequency or transient interference will falsely operate a d.c. track relay.

These standard limits and methodologies will facilitate the demonstration of compatibility with d.c. track circuits for rolling stock, as part of a Route Acceptance Safety Case.

I.2 Scope

This document covers a.c. immune d.c. track circuits installed in single rail mode on a.c. electrified Network Rail Controlled Infrastructure.

The methodology considers both intact and faulty infrastructure. Limits are established for both continuous welded rail (CWR) and jointed rail (JR) track types.

The maximum permissible levels (i.e. the limits) of d.c. interference current in the tracks which will not cause a wrong side failure (WSF) (i.e. false operation of the relay), have been derived, together with guidance on appropriate apportionment factors.

Standard train current and axle to axle voltage steady state limits are defined for the predominant types of a.c. immune d.c. track relay currently in use. Appropriate limits for other relay types (except BR966F9) may be determined by scaling with respect to specified operating voltages.

2 SOURCE DOCUMENTS

2.1 Railway Group Standards

- 2.1.1 GK/RT0011, "Train Detection", Issue 3, August 2000
- 2.1.2 GK/GN0611, "Train Detection", Issue 1, August 2000

2.2 Network Rail Company Standards

- 2.2.1 RT/E/S/11752, "Train Detection", Issue 2, August 2001
- 2.2.2 RT/E/PS/11755, "D.C. Track Circuits", Issue 1, December 2000
- 2.2.3 RT/E/S/21078, "Design of Return Conductor Systems for A.C. Electrified Lines", Issue 1, March 1998
- 2.2.4 RT/E/S/21085, "Design of Earthing and Bonding Systems for 25kV Electrified Lines", Issue 1, October 1998

2.3 Previous Safety Case Documentation

- 2.3.1 "Class 390, DC Track Circuit Compatibility", 10150.1250, Revision B [ALSTOM], 26/4/02
- 2.3.2 "Class 390, DC Track Circuit Risk Assessment", 10150.1238, Revision B [ALSTOM], 26/4/02
- 2.3.3 "Class 334, DC Track Circuit Compatibility, 6-Car Train", 10149.1230, Revision C [ALSTOM], 28/11/00
- 2.3.4 "Class 373 Support – Reed Track Circuits Infrastructure Modifications for Increased Immunity", STG-D-EPS(373)-RP-29, Issue 1.1, [AEA] March 1997
- 2.3.5 "AC Immune d.c. Track Relay Tests", LR ES 009, [BRR] December 1993
- 2.3.6 "DC Track Circuit Relays, Static Tests" STG-D-RFD(092)-RP-8, Issue 1, [BRR] August 1995
- 2.3.7 "Characterisation of DC Track Relays (Report 98097) – Normalisation of Frequency Response Results", C2180-1, [WSAR] 14/9/98
- 2.3.8 "Investigation on Switching Frequency Monitor Tripping Under Conditions of Icy OHL", F0095.56.8-2, Issue 1.1, [Siemens] 6/3/01
- 2.3.9 "Class 333 4QC DC Monitor Tripping", F0095.56.8/VT872.276.Ne, [Siemens] 21/3/01
- 2.3.10 "Response to 50 Hz of DC Track Circuit Relays", Bombardier Transportation Technical Report 3EER300002-1771 revision, 3/7/01.

2.4 Industry Data Initiative Documentation

- 2.4.1 I737001-C-TR-001 – “Strategy for the Development and Acceptance of ECPs”, Version 1.0, 18/11/99.
- 2.4.2 I737001-L-TR-001 – “Identification of Sources”, Version 0.1, 3/11/99.
- 2.4.3 SFCG03-DCTC – “Shortform Compatibility Guide, Volume 3 (D.C. Track Circuits)”, Version 1.0, 18/5/00
- 2.4.4 I737001-C-TR-003 – “DC Track Circuit Susceptibility Limits Model”, Version 0.4, 9/6/00
- 2.4.5 I737001-C-TR-004 – “DC Track Circuit Simplified Steady State Model”, Version 0.1, 14/6/00

2.5 Other Documents

- 2.5.1 “Summary of Results DCMOD13a to DCMOD20a”, Filename: ‘Single and Double Track Results_Version b.xls’, [Railtrack] 10/6/02
- 2.5.2 “Models – DCMOD21a and DCMOD22a”, Filename: ‘Four Track dc results – Jointed Rail_Version a.xls’, [Railtrack] 6/6/02
- 2.5.3 BR939A, “Specification for miniature tractive armature AC immune DC neutral track relay, plug-in type for railway signalling purposes”, September 1971
- 2.5.4 BR966, “Specification for miniature tractive armature relays, plug-in type for railway signalling purposes”, August 1981
- 2.5.5 BR933A, “Specification for miniature tractive armature AC immune DC slow pick-up neutral line relay, plug-in type for railway signalling purposes”, September 1971
- 2.5.6 “Railway Safety - HM Chief Inspecting Officer of Railways’ Annual Report on the safety record of the railways in Great Britain during 1995/6” HSE Books.
- 2.5.7 prEN50328, “Compatibility between Rolling Stock and Train Detection Systems”, Final Draft 1998

3 D.C. TRACK CIRCUIT BACKGROUND

3.1 D.C. Track Circuit Operation

D.c. track circuits provide a simple and reliable track circuit for use on non d.c. electrified lines. In their basic form, power is applied to the rails at one end of the section and is transmitted via the rails to the d.c. track relay at the other end.

Two principal installation configurations are employed – double rail, where both rails are fitted with IRJs (Insulated Rail Joints) at each end of the section to keep adjacent track circuits electrically separate, and single rail, where only one of the rails is fitted with IRJs at either end of the section, and the other rail is thus electrically common

with the adjacent track circuit. To permit the flow of traction return current, d.c. track circuits on a.c. electrified lines are exclusively of the single rail type.

The d.c. power supply is obtained from a transformer/rectifier or a battery, and is connected through a feed resistor to the rails. This resistor limits the current drawn from the power supply when the track circuit is shunted by the train, and permits the track circuit sensitivity to be adjusted. The performance of d.c. track circuits is determined by:

- the shunt resistance presented by the train;
- the drop away rail voltage (to tolerate d.c. interference and rail contamination);
- the minimum ballast resistance at which the track circuit remains functional;
- the need, or otherwise, for a.c. immunity.

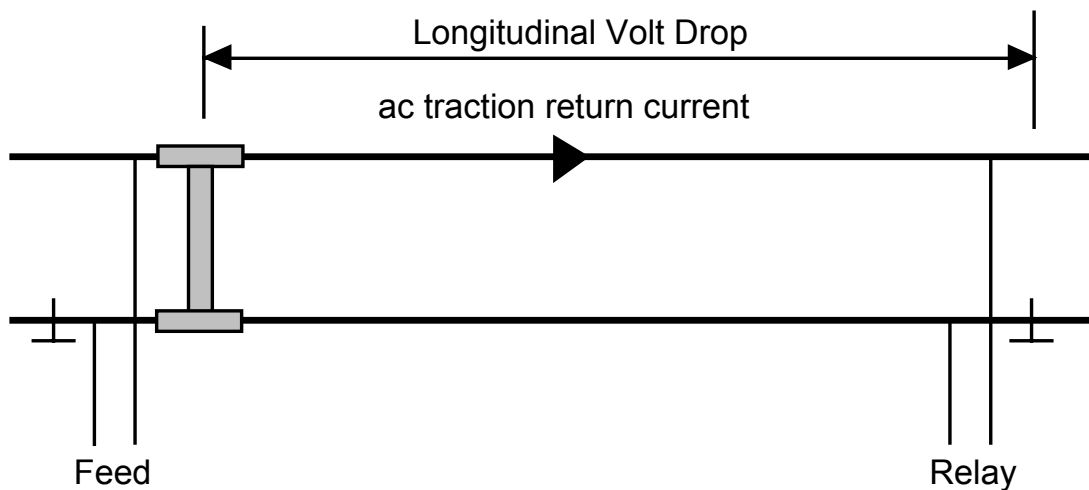


Figure 1: Single Rail D.C. Track Circuit in A.C. Electrified Areas

In a.c. electrified areas, a.c. immunity is aided by connecting a choke at the feed end of the circuit, whilst the track relay itself is fitted with a slug to prevent a.c. magnetisation.

A number of variants of d.c. track circuit have been employed over the years each having different electrical parameters. However, of particular interest in determining permissible interference levels, are the relay characteristics. Available data for various relay types are provided in table 2.

The improved shunt characteristic of track circuits fitted with feed-end relays is not specifically considered in this document.

Detailed descriptions of d.c. track circuits and their operation and installation are given in Network Rail Line Product Specification RT/E/PS/11755 (ref. 2.2.2).

3.2 Interference with D.C. Track Circuits

The main interference mechanism of interest is where the d.c. content of the a.c. traction return current will produce a voltage drop presented across the track relay by the train shunt (see Figure 1). To create the hazard from an interference current, the d.c. component of the interference current that flows in the rails must develop at least the drop-away voltage of the relay over the length of track between the train and the track circuit relay.

Thus the main factors to be considered, when assessing the susceptibility of a d.c. track circuit system, are:

- the sensitivity of the track circuit relay, which depends on the type of relay;
- track resistance per unit length, which depends on the type of track (CWR or JR) and the track layout;
- the effective length of the return path (track circuit length, plus any run round path due to faults).

As the susceptibility level is roughly linearly proportional to any and all of these factors, it may prove unreasonable to consider the “worst case” (i.e. the most sensitive track relay and the longest track circuit on the highest-resistance track) as the generally applicable limit, as this may be significantly worse than that applicable to any individual track circuit on a given route.

Where routes include both sections of jointed rail and low voltage relays (e.g. other than to BR939A or BR966F9, refs. 2.5.3, 2.5.4), a more reasonable approach is for the maximum track circuit lengths and resistances to be determined, for the low voltage relays in use on any jointed rail (JR) fitted sections of route; the permissible interference current then being calculated for each actual combinations of relay type and return path resistance.

The influence of infrastructure faults (a break in the traction return rail, or a fault electrically equivalent to that), which increase the effective length of traction return path, must also be considered.

The contribution of other trains in the vicinity to the interference current is allowed

for by including a suitable apportionment factor.

4 INFRASTRUCTURE ASSUMPTIONS

The main infrastructure assumptions that have been used in the d.c. track circuit analysis (refs. 2.5.1, 2.5.2) are listed below.

4.1 Parameters

Parameter	Value	Data source
Ballast resistance	Infinite	Refs. 2.5.1, 2.5.2
Structure to earth resistance	20 Ω	Ref. 2.3.1
Feeder station return to earth resistance at end of modelled section	1 Ω	Overall return resistance to earth <1 Ω , Ref. 2.2.3
Train shunt resistance	100.1m Ω	Refs. 2.5.1, 2.5.2
Cross-bond cable resistance	1m Ω	Refs. 2.5.1, 2.5.2
Return wire resistance (19/4.22mm Al)	117m Ω /km	Refs. 2.5.1, 2.5.2
Earth wire resistance (19/3.25mm Al)	198m Ω /km	Refs. 2.5.1, 2.5.2
Rail resistance (CWR)	35 m Ω /km	Ref. 2.2.1
Rail resistance (60ft JR – jointed rail)	250 m Ω /km	Ref. 2.2.1 (assumes 4m Ω per joint)
Cross-bond spacing (return conductor)	780m	750m max. specified in ref. 2.2.4
Cross-bond spacing (rail return)	540m	500m max. specified in ref. 2.2.4
Earth Wire bond spacing (rail return / earth-wire)	780m	800m max. specified in ref. 2.2.4
MPC (Mid Point Connection) spacing	3.12km	3.2km average specified in ref. 2.2.3
Track feed (9 Ω relay)	6V / 7.2 Ω	Refs. 2.5.1, 2.5.2
Track feed (20 Ω relay)	10V / 7.2 Ω	Refs. 2.5.1, 2.5.2

Table 1 - Parameter Values Applied in D.C. Track Circuit Analysis

Inductances have been omitted, as the modelling is steady state and time independent.

4.2 Other Infrastructure Assumptions

The following assumptions have been made and are listed together with a statement as to whether they are pessimistic (generally meaning worst case), realistic (generally meaning not worst case) or neutral with regard to their influence on the interference

current limit which results.

Although the assumptions taken individually are either neutral, realistic or reasonably pessimistic, they can be extremely pessimistic when combined.

- The interference limits are derived on the basis of specified track relay operating voltages, rather than specified operating currents. Nominal relay coil resistances are assumed for the purpose of determining relay operating voltages. [Realistic]
- Relays which are pin-code compatible with BR939A or BR966 relays have similar susceptibility characteristics to those relays which conform strictly to those specifications (refs. 2.5.3, 2.5.4). [Realistic]
- Nominal relay coil resistances are assumed for the purpose of deriving the potential impressed across it by the interference current. [Realistic]
- Feed through from the track circuit feed is calculated from nominal values, and does not therefore require specific consideration within an apportionment factor. [Neutral]
- The train is modelled as a single axle shunting the rails, occupying only the track circuit in question. (It is noted that at certain locations and/or with longer train formations, more than one track circuit may be occupied providing additional train protection. Additionally, current return path lengths are reduced with longer trains thereby reducing sensitivity). [Pessimistic]
- Train interference is assumed to be a perfect d.c. current source. [Pessimistic]
- All electrification bonding complies with ref. 2.2.4. [Realistic]
- Return Conductor arrangements comply with ref. 2.2.3 and Appendix C of 2.2.4. [Realistic]
- Earth Wire arrangements comply with Appendix C of 2.2.4. [Realistic]
- OHLE structures are sited on average at 60m intervals. [realistic]
- Each OHLE structure has a 20Ω d.c. resistance to earth. [realistic]
- Other than in earth-wire areas, each OHLE structure is bonded to the nearest return rail. [realistic]
- The resistance of OHLE structure to return rail bonds is not considered as this is very low in comparison to the structure to earth resistance. [realistic]
- In earth-wire areas, each OHLE structure is bonded to the earth-wire. [realistic]
- Return wire is 19/4.22mm Aluminium. [Realistic]
- Earth wire is 19/3.25mm Aluminium. [Realistic]
- It is assumed that no traction return current flows in the signal rails of adjacent track circuits. [Pessimistic]
- Parallel tracks all have single rail traction return (even if not fitted with d.c. track circuits). [Pessimistic]
- In multiple track areas, cross bonds between parallel tracks are assumed to occur at the same route mileage position.[Neutral]

- The position of cross bonds are assumed to be such that they have the worst effect on the track circuit relay. [Pessimistic]
- The position of the train is assumed to be such that it has the worst effect on the track circuit relay, i.e. just beyond the transmitter in the direction of the relay. [Pessimistic]
- When track maintenance degrades the infrastructure for temporary periods the infrastructure controller will implement mitigation measures to ensure that track circuits are not made significantly more susceptible. For example, the singling of double CWR track with temporary JR makes the track circuit more susceptible. A mitigating factor would be to ensure the length of JR is significantly greater than 60ft (standard JR length) during the installation. [Realistic]

4.3 Infrastructure Failure Modes

- Only single failures (e.g. not a coincident broken rail and disconnected cross bond) are considered.
- A single broken traction rail (or electrically equivalent series bonding failure) is considered, this being the most significant failure mode. A single broken cross bond will result in interference limits that tend towards the single track case.
- The position of cross bonds, mid point connections (MPCs) and earth wire connections, are assumed to be such that they have the worst effect on the track circuit relay (e.g. an effective cross bond is positioned just beyond the rail break in the direction of the relay).

- The position of any rail break is assumed to be such as to have the worst effect on track circuit relay, i.e. just beyond the train in the direction of the relay.
- Failures of the signal (insulated) rail (or electrically equivalent series bonding failure) are not considered, as in a previous study applicable to all single rail track circuits of audio frequency or less (ref. 2.3.4), it has been demonstrated that the allowable interference currents for these revealed failure modes are orders of magnitude greater than for the unrevealed failure modes identified above.

5 DERIVATION OF LIMITS (STEADY STATE)

The limits for d.c. and low frequency interference on the a.c. electrified railway have been calculated in refs. 2.5.1 & 2.5.2.

These show the level of interference current required to produce a voltage across the track relay equal to its minimum drop away voltage.

Eight different a.c. immune d.c. track relay types, which account for the majority of those in use, have been considered. The following table summarises key data for these relay types. Specified values are given based on the parameters given in the relay specifications, or lowest measured values, where specifications have not been obtained:

Relay Type	Nominal Coil Resistance	Minimum Drop-Away Voltage	Minimum Pick-Up Voltage	Maximum Length of Track Circuit (metres) #1
BR939A	20 Ω	1.100 V	1.620 V	1000
BR966 F2	9 Ω	0.730 V	1.080 V	680
BR966 F9	60 Ω	1.900 V	2.800 V	350
B1	9 Ω	0.531 V	0.740 V	600
CAA02	9 Ω	0.470 V	0.710 V	600
NT2A	9 Ω	0.469 V	0.600 V	600
D6	9 Ω	0.427 V	0.550 V	600
CAA04	9 Ω	0.290 V	0.440 V	600

#1 Maximum track circuit lengths are proposed in ref. 2.2.2: there are however significant numbers of non-compliant track circuits

Table 2 – D.C. Track Relay Characteristics

The most sensitive d.c. track relay which is used in a.c. electrified areas is the CAA04, which has a minimum specified drop-away voltage of 0.29V

The results of the calculations define the levels of d.c. interference required to produce a voltage across the relay equal to its minimum drop away voltage. It should be noted that there are no specific apportionment factors built into these calculations.

Where specific infrastructure conditions are known then the appropriate value can be obtained from the figures and tables detailed in sections 5.1 to 5.6 below and used as part of the design process and/or electrical compatibility assessment. A suitable apportionment factor must be applied to them; section 10 refers.

Where the specific infrastructure conditions are not known then the worst case shall be assumed, again with a suitable apportionment factor applied.

The results of the analysis presented in refs. 2.5.1 & 2.5.2 are summarised below.

5.1 Single Track Railway: Intact Infrastructure

For a single track railway (with no cross bonding present) the analysis presented in ref. 2.5.1 is summarised in figs. 2, 3 and 4 together with tables 3, 4, 5 and 6 below (the tables contain the data from which the graphs are plotted).

It should be noted that only the results for the relay types BR939A, CAA02 and CAA04 are summarised in the figures which are plots of minimum interference current against track circuit length.

Figs. 2, 3 and 4 below each contain four plots. Each plot relates to a specific set of calculations conducted in ref. 2.5.1 and is identified by way of a reference to a specific calculation model. The calculations carried out by each of the four models are detailed below:

- DCMOD13a: single track railway, jointed rail, booster sections with return conductor.
- DCMOD15a: single track railway, CWR, booster sections with return conductor.
- DCMOD17a: single track railway, jointed rail with rail return and earth-wire.
- DCMOD19a: single track railway, CWR with rail return and earth-wire.

Each of the four models has been used to calculate limits for both intact infrastructure (this section) and failed infrastructure (section 5.2 below).

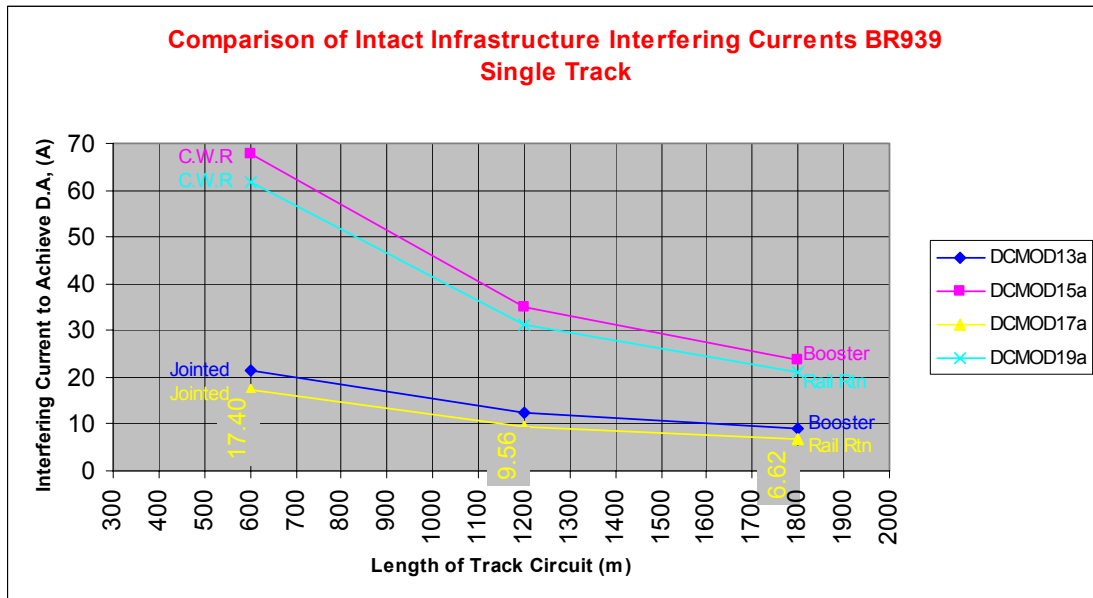


Fig. 2 – Single Track Intact Infrastructure Interference Current Limits BR939A

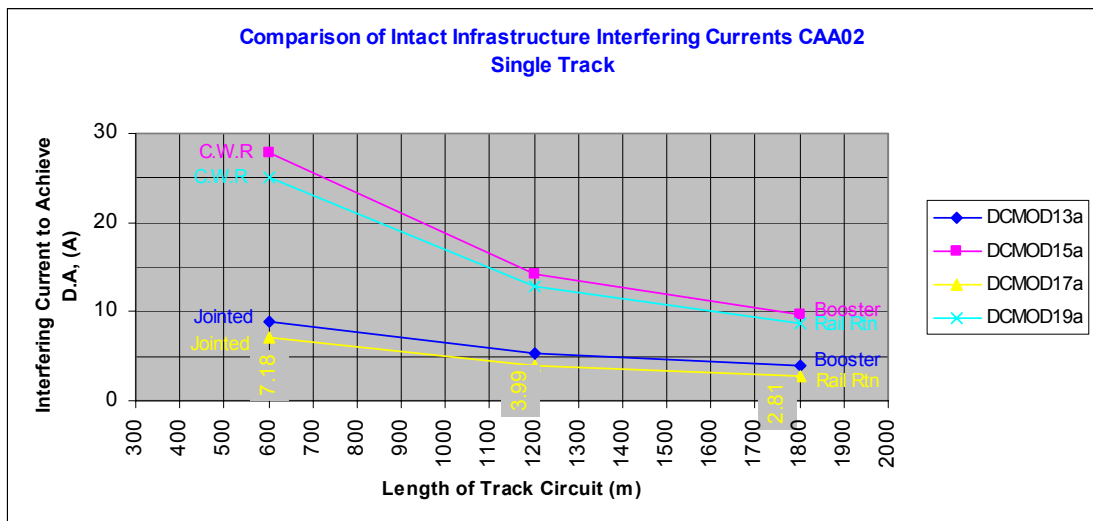


Fig. 3 – Single Track Intact Infrastructure Interference Current Limits CAA02

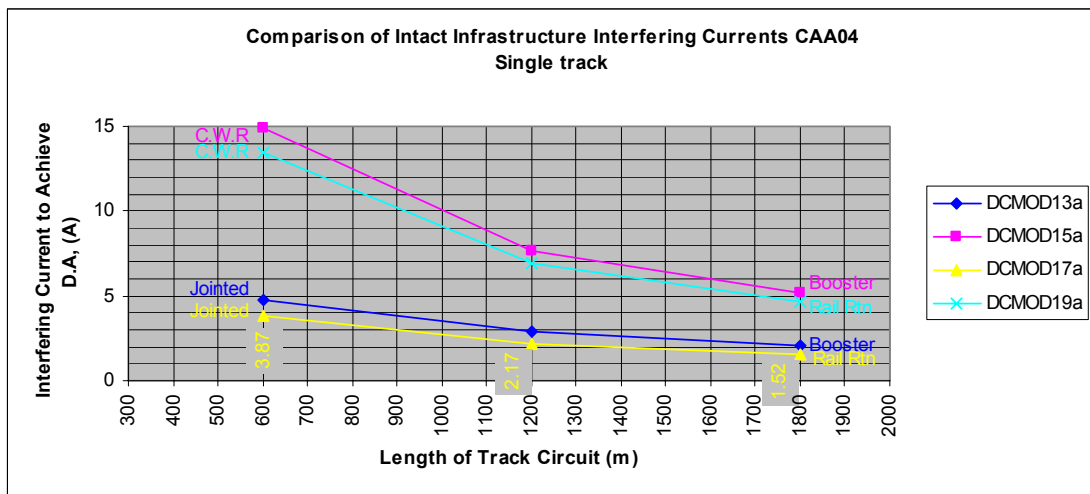


Fig. 4 – Single Track Intact Infrastructure Interference Current Limits CAA04

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	21.35	12.52	9.02
BR966 F2	9	0.730	14.71	8.77	6.39
BR966 F9	60	1.900	Not Calculated > 10A		
BI	9	0.531	10.22	6.09	4.44
CAA02	9	0.470	8.84	5.28	3.85
NT2A	9	0.469	8.82	5.26	3.84
D6	9	0.427	7.87	4.7	3.43
CAA04	9	0.290	4.77	2.85	2.09

Table 3 – Single Track Intact Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	67.77	34.89	23.64
BR966 F2	9	0.730	46.48	23.77	16.12
BR966 F9	60	1.900	Not Calculated > 35A		
BI	9	0.531	32.23	16.49	11.18
CAA02	9	0.470	27.88	14.26	9.67
NT2A	9	0.469	27.79	14.22	9.64
D6	9	0.427	24.78	12.69	8.60
CAA04	9	0.290	14.93	7.67	5.20

Table 4 – Single Track Intact Infrastructure (CWR, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	17.40	9.56	6.62
BR966 F2	9	0.730	11.95	6.64	4.66
BR966 F9	60	1.900	Not Calculated > 10A		
BI	9	0.531	8.30	4.61	3.24
CAA02	9	0.470	7.18	3.99	2.81
NT2A	9	0.469	7.16	3.98	2.80
D6	9	0.427	6.39	3.56	2.50
CAA04	9	0.290	3.87	2.17	1.52

Table 5 – Single Track Intact Infrastructure (Jointed Rail, Rail Return and Earth-Wire) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	61.72	31.29	21.03
BR966 F2	9	0.730	41.90	21.31	14.32
BR966 F9	60	1.900	Not Calculated > 30A		
BI	9	0.531	29.05	14.78	9.94
CAA02	9	0.470	25.12	12.78	8.59
NT2A	9	0.469	25.05	12.75	8.57
D6	9	0.427	22.34	11.37	7.64
CAA04	9	0.290	13.49	6.88	4.63

Table 6 – Single Track Intact Infrastructure (CWR, Rail Return and Earth-Wire)
Calculated Interference Current

5.2 Single Track Railway: Failed Infrastructure

For a single track railway with failed infrastructure (worst case rail break giving maximum length run round path for interference current) the analysis presented in ref. 2.5.1 is summarised in figs. 5, 6 and 7 together with tables 7, 8, 9 and 10 below (the tables contain the data from which the graphs are plotted).

It should be noted that only the results for the relay types BR939A, CAA02 and CAA04 are summarised in the figures which are plots of minimum interference current against track circuit length.

Figs. 5, 6 and 7 below each contain four plots. Each plot relates to a specific set of calculations conducted in ref. 2.5.1 and is identified by way of a reference to a specific calculation model. The calculations carried out by each of the four models are detailed below:

- DCMOD13a: single track railway, jointed rail, boosters section with return conductor.
- DCMOD15a: single track railway, CWR, booster sections with return conductor.
- DCMOD17a: single track railway, jointed rail with rail return and earth-wire.
- DCMOD19a: single track railway, CWR with rail return and earth-wire.

Each of the four models has been used to calculate limits for both failed infrastructure (this section) and intact infrastructure (section 5.1 above).

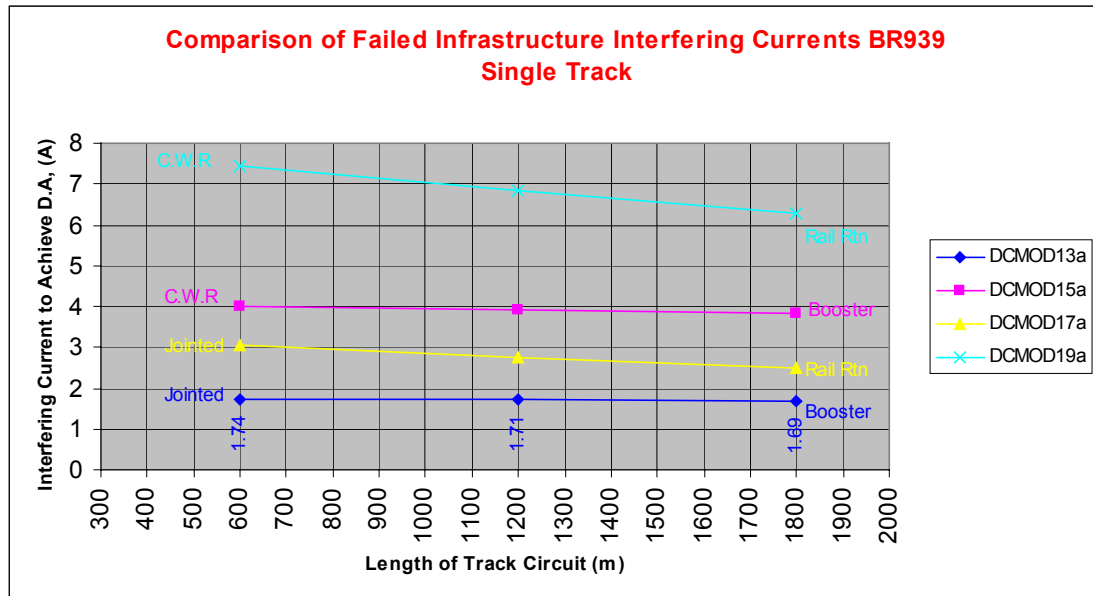


Fig. 5 – Single Track Failed Infrastructure Interference Current Limits BR939A

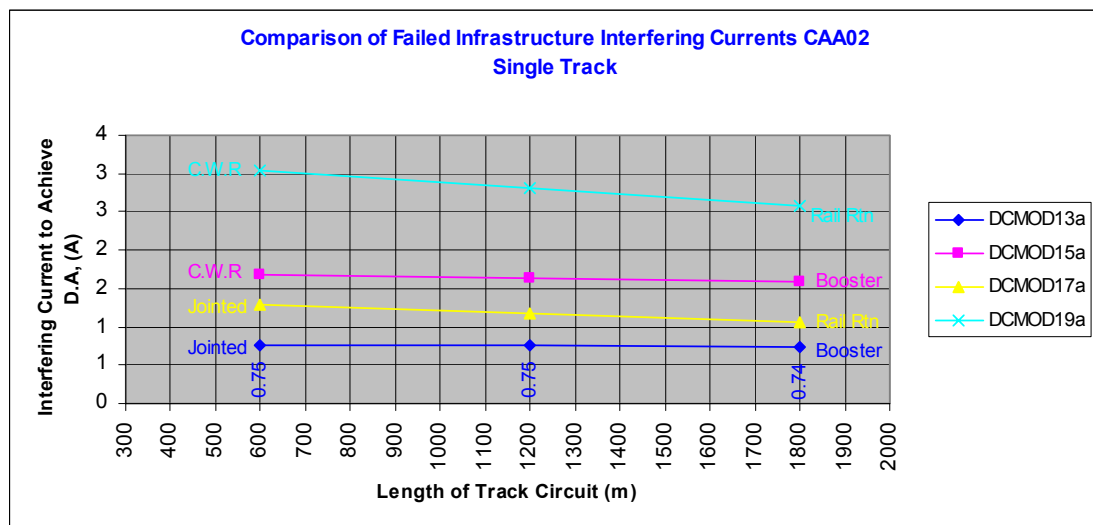


Fig. 6 – Single Track Failed Infrastructure Interference Current Limits CAA02

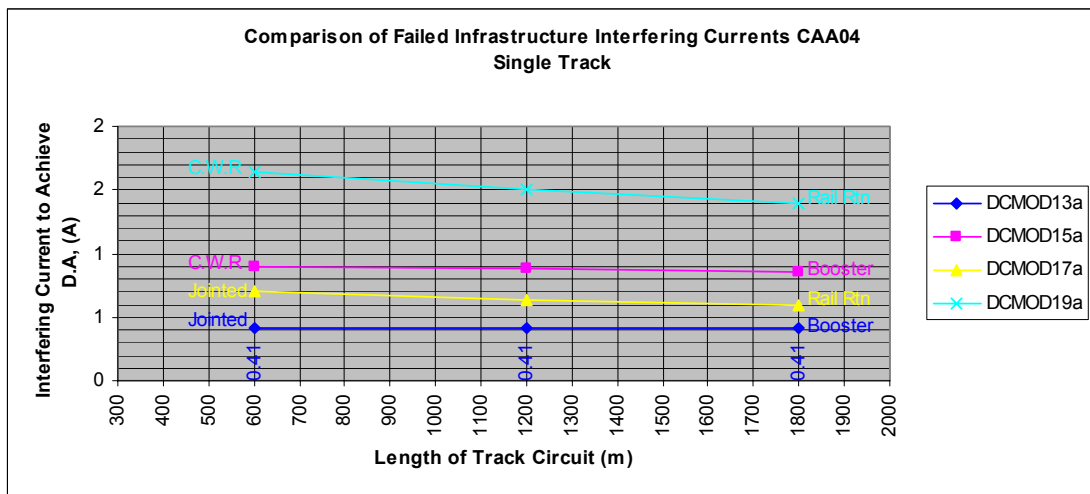


Fig. 7 – Single Track Failed Infrastructure Interference Current Limits CAA04

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	1.74	1.71	1.69
BR966 F2	9	0.730	1.24	1.24	1.22
BR966 F9	60	1.900	Not Calculated > 2.5A		
BI	9	0.531	0.86	0.87	0.85
CAA02	9	0.470	0.75	0.75	0.74
NT2A	9	0.469	0.75	0.75	0.74
D6	9	0.427	0.67	0.67	0.66
CAA04	9	0.290	0.41	0.41	0.41

Table 7 – Single Track Failed Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	4.02	3.92	3.81
BR966 F2	9	0.730	2.78	2.71	2.64
BR966 F9	60	1.900	Not Calculated > 6A		
BI	9	0.531	1.93	1.88	1.84
CAA02	9	0.470	1.67	1.63	1.59
NT2A	9	0.469	1.67	1.63	1.58
D6	9	0.427	1.49	1.45	1.41
CAA04	9	0.290	0.9	0.88	0.86

Table 8 – Single Track Failed Infrastructure (CWR, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	3.04	2.74	2.49
BR966 F2	9	0.730	2.13	1.94	1.77
BR966 F9	60	1.900	Not Calculated > 4A		
BI	9	0.531	1.48	1.35	1.24
CAA02	9	0.470	1.28	1.17	1.07
NT2A	9	0.469	1.28	1.17	1.07
D6	9	0.427	1.14	1.04	0.96
CAA04	9	0.290	0.70	0.64	0.59

Table 9 – Single Track Failed Infrastructure (Jointed Rail, Rail Return and Earth-Wire) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	7.44	6.82	6.30
BR966 F2	9	0.730	5.08	4.68	4.31
BR966 F9	60	1.900	Not Calculated > 10A		
BI	9	0.531	3.53	3.25	3.00
CAA02	9	0.470	3.05	2.81	2.59
NT2A	9	0.469	3.04	2.80	2.58
D6	9	0.427	2.71	2.50	2.31
CAA04	9	0.290	1.64	1.51	1.40

Table 10 – Single Track Failed Infrastructure (CWR, Rail Return and Earth-Wire)
Calculated Interference Current

5.3 Double Track Railway: Intact Infrastructure

For a double track railway the analysis presented in ref. 2.5.1 is summarised in figs. 8, 9 and 10 together with tables 11, 12, 13 and 14 below (the tables contain the data from which the graphs are plotted).

It should be noted that only the results for the relay types BR939A, CAA02 and CAA04 are summarised in the figures which are plots of minimum interference current against track circuit length.

Figs. 8, 9 and 10 below each contain four plots. Each plot relates to a specific set of calculations conducted in ref. 2.5.1 and is identified by way of a reference to a specific calculation model. The calculations carried out by each of the four models are detailed below:

- DCMOD14a: double track railway, jointed rail, booster sections with return conductor.
- DCMOD16a: double track railway, CWR, booster sections with return conductor.
- DCMOD18a: double track railway, jointed rail with rail return.
- DCMOD20a: double track railway, CWR with rail return.

Each of the four models has been used to calculate limits for both intact infrastructure (this section) and failed infrastructure (section 5.4 below).

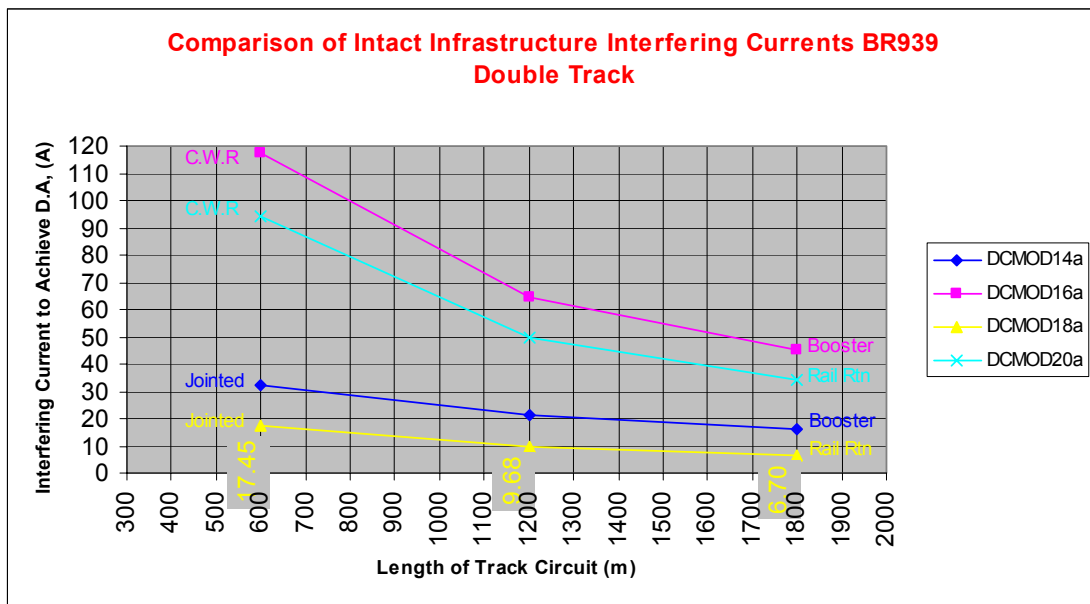


Fig. 8 – Double Track Intact Infrastructure Interference Current Limits BR939A

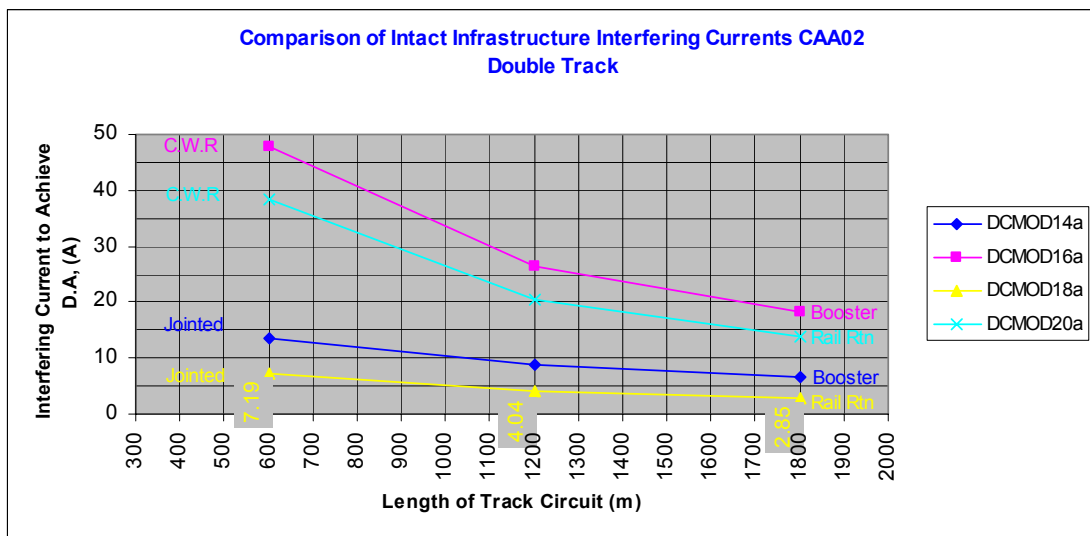


Fig. 9 – Double Track Intact Infrastructure Interference Current Limits CAA02

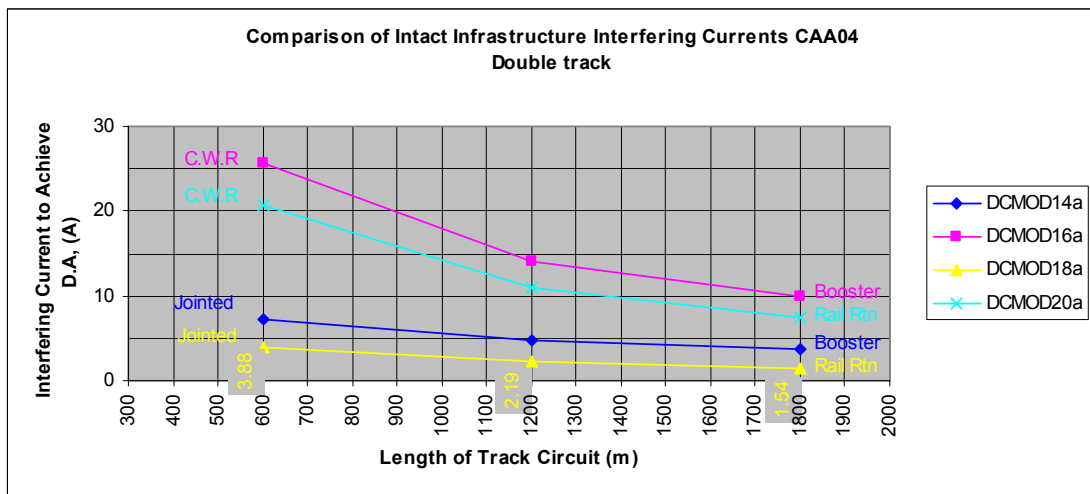


Fig. 10 – Double Track Intact Infrastructure Interference Current Limits CAA04

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	32.56	21.17	15.95
BR966 F2	9	0.730	22.36	14.71	11.23
BR966 F9	60	1.900	Not Calculated > 25A		
BI	9	0.531	15.52	10.22	7.80
CAA02	9	0.470	13.42	8.85	6.75
NT2A	9	0.469	13.39	8.83	6.74
D6	9	0.427	11.95	7.88	6.01
CAA04	9	0.290	7.24	4.79	3.66

Table 11 – Double Track Intact Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	117.57	64.82	44.86
BR966 F2	9	0.730	79.66	43.93	30.33
BR966 F9	60	1.900	Not Calculated > 70A		
BI	9	0.531	55.23	30.47	21.05
CAA02	9	0.470	47.78	26.36	18.17
NT2A	9	0.469	47.62	26.28	18.16
D6	9	0.427	42.47	23.44	16.20
CAA04	9	0.290	25.59	14.15	9.87

Table 12 – Double Track Intact Infrastructure (CWR, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	17.45	9.68	6.70
BR966 F2	9	0.730	11.97	6.72	4.73
BR966 F9	60	1.900	Not Calculated > 10A		
BI	9	0.531	8.31	4.67	3.29
CAA02	9	0.470	7.19	4.04	2.85
NT2A	9	0.469	7.17	4.03	2.84
D6	9	0.427	6.40	3.60	2.54
CAA04	9	0.290	3.88	2.19	1.54

Table 13 – Double Track Intact Infrastructure (Jointed Rail, Rail Return) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	94.35	49.97	33.92
BR966 F2	9	0.730	64.01	33.86	23.07
BR966 F9	60	1.900	Not Calculated > 50A		
BI	9	0.531	44.39	23.49	16.00
CAA02	9	0.470	38.37	20.30	13.84
NT2A	9	0.469	38.27	20.26	13.80
D6	9	0.427	34.13	18.07	12.31
CAA04	9	0.290	20.63	10.95	7.44

Table 14 – Double Track Intact Infrastructure (CWR, Rail Return) Calculated Interference Current

5.4 Double Track Railway: Failed Infrastructure

For a double track railway with failed infrastructure (worst case rail break giving maximum length run round path for interference current) the analysis presented in ref. 2.5.1 is summarised in figs. 11, 12 and 13 together with tables 15, 16, 17 and 18 below (the tables contain the data from which the graphs are plotted).

It should be noted that only the results for the relay types BR939A, CAA02 and CAA04 are summarised in the figures which are plots of minimum interference current against track circuit length.

Figs. 11, 12 and 13 below each contain four plots. Each plot relates to a specific set of calculations conducted in ref. 2.5.1 and is identified by way of a reference to a specific calculation model. The calculations carried out by each of the four models are detailed below:

- DCMOD14a: double track railway, jointed rail, booster sections with return conductor.
- DCMOD16a: double track railway, CWR, booster sections with return conductor.
- DCMOD18a: double track railway, jointed rail with rail return.
- DCMOD20a: double track railway, CWR with rail return.

Each of the four models has been used to calculate limits for both failed infrastructure (this section) and intact infrastructure (section 5.3 above).

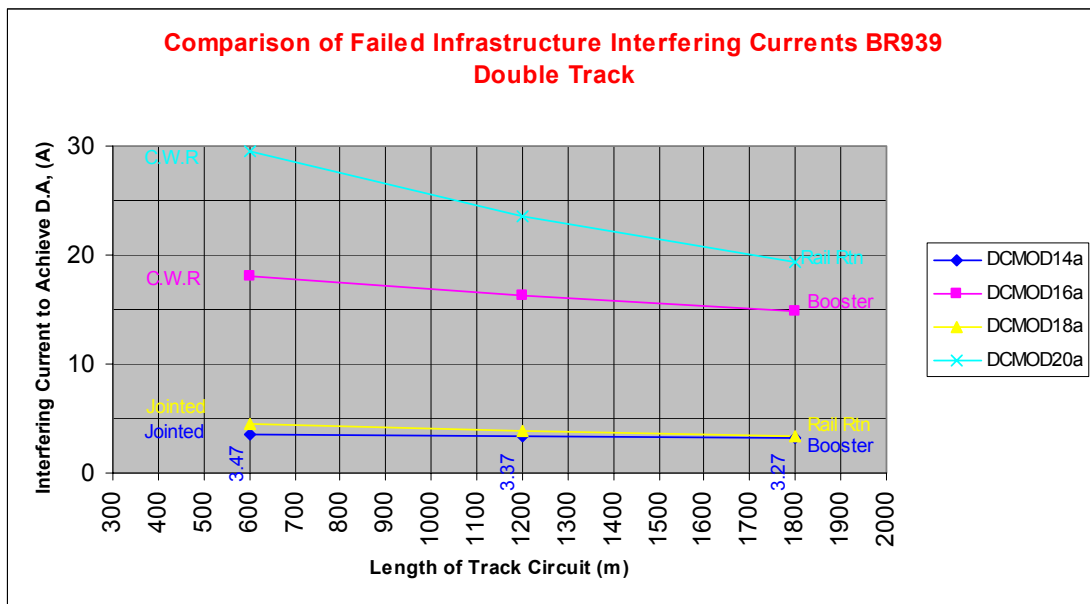


Fig. 11 - Double Track Failed Infrastructure Interference Current Limits BR939A

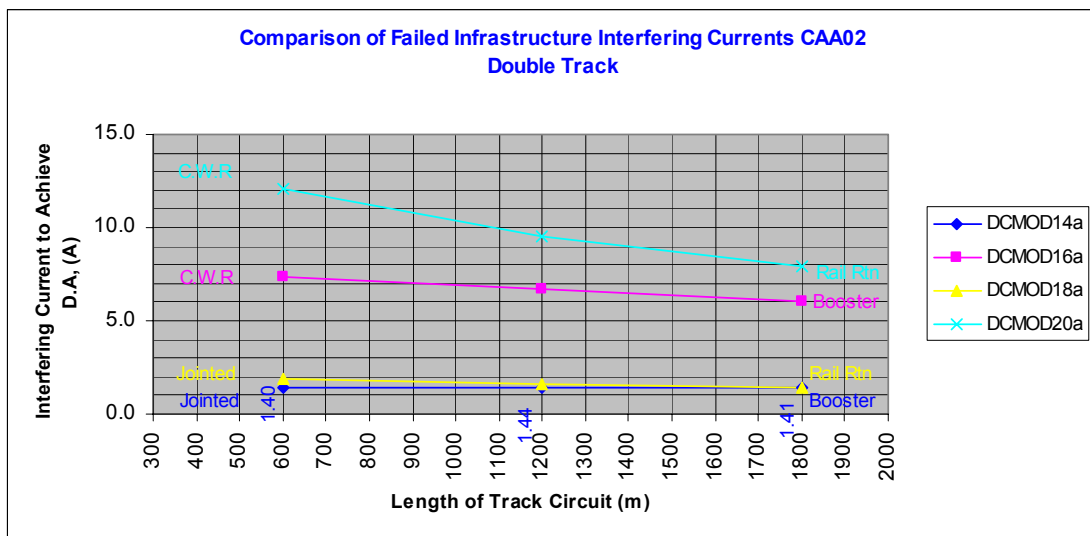


Fig. 12 - Double Track Failed Infrastructure Interference Current Limits CAA02

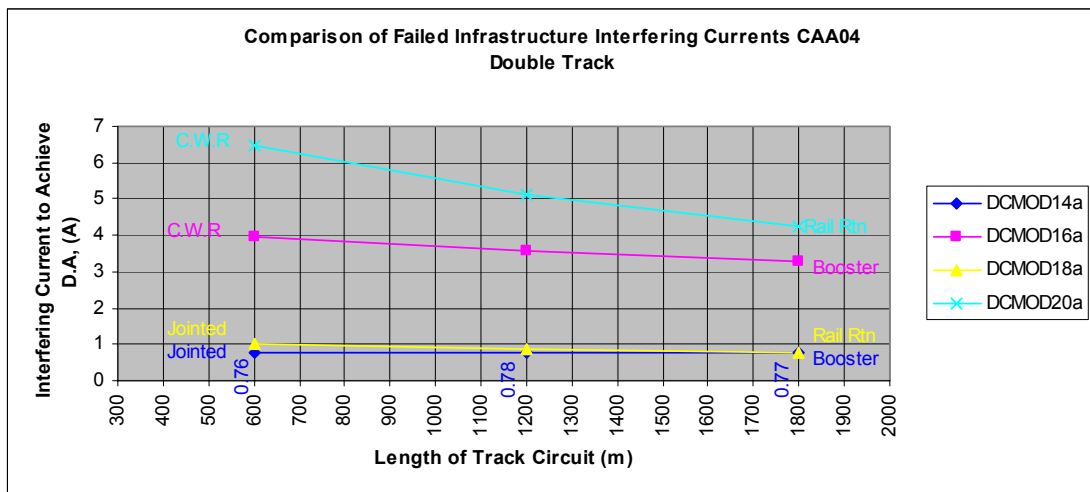


Fig. 13 - Double Track Failed Infrastructure Interference Current Limits CAA04

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	3.47	3.37	3.27
BR966 F2	9	0.730	2.33	2.39	2.34
BR966 F9	60	1.900	Not Calculated > 5A		
BI	9	0.531	1.62	1.62	1.62
CAA02	9	0.470	1.40	1.40	1.40
NT2A	9	0.469	1.40	1.40	1.40
D6	9	0.427	1.25	1.25	1.25
CAA04	9	0.290	0.76	0.76	0.76

Table 15 – Double Track Failed Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	18.03	16.28	14.78
BR966 F2	9	0.730	12.29	11.10	10.10
BR966 F9	60	1.900	Not Calculated > 20A		
BI	9	0.531	8.53	7.70	7.01
CAA02	9	0.470	7.37	6.66	6.06
NT2A	9	0.469	7.36	6.64	6.05
D6	9	0.427	6.56	5.92	5.39
CAA04	9	0.290	3.98	3.58	3.27

Table 16 – Double Track Failed Infrastructure (CWR, Booster Sections, Return Conductor) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	4.46	3.85	3.37
BR966 F2	9	0.730	3.11	2.71	2.40
BR966 F9	60	1.900	Not Calculated > 5A		
BI	9	0.531	2.16	1.88	1.87
CAA02	9	0.470	1.87	1.63	1.45
NT2A	9	0.469	1.86	1.63	1.45
D6	9	0.427	1.66	1.45	1.29
CAA04	9	0.290	1.01	0.89	0.79

Table 17 – Double Track Failed Infrastructure (Jointed Rail, Rail Return) Calculated Interference Current

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	29.59	23.47	19.41
BR966 F2	9	0.730	20.08	15.95	13.21
BR966 F9	60	1.900	Not Calculated > 30A		
BI	9	0.531	13.93	11.06	9.16
CAA02	9	0.470	12.04	9.57	7.93
NT2A	9	0.469	12.01	9.54	7.90
D6	9	0.427	10.72	8.51	7.05
CAA04	9	0.290	6.49	5.14	4.25

Table 18 – Double Track Failed Infrastructure (CWR, Rail Return) Calculated Interference Current

5.5 Four Track Railway: Intact Infrastructure

For a four track railway with intact infrastructure the analysis presented in ref. 2.5.2 is summarised in table 19 below.

The calculations were carried out via one model as detailed below:

- DCMOD21a: four track railway, jointed rail, booster sections with return conductor.

No calculations have been performed for the CWR track as these will provide higher susceptibility figures.

The model has been used to calculate limits for both intact infrastructure (this section) and failed infrastructure (section 5.6 below).

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	42.04	30.13	23.19
BR966 F2	9	0.730	28.92	20.94	16.73
BR966 F9	60	1.900	Not Calculated > 35A		
BI	9	0.531	20.08	14.55	11.63
CAA02	9	0.470	17.37	12.59	10.08
NT2A	9	0.469	17.32	12.56	10.04
D6	9	0.427	15.45	11.21	8.97
CAA04	9	0.290	9.35	6.82	5.44

Table 19 – Four Track Intact Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

5.6 Four Track Railway: Failed Infrastructure

For a four track railway with failed infrastructure the analysis presented in ref. 2.5.2 is summarised in table 20 below.

The calculations were carried out via one model as detailed below:

- DCMOD21a: four track railway, jointed rail, booster sections with return conductor.

No calculations have been performed for the CWR track or for rail return conditions, as these will provide higher susceptibility figures.

The model has been used to calculate limits for both intact infrastructure (this section) and intact infrastructure (section 5.5 above).

Relay Type	Relay Coil Resistance (Ω)	Drop Away Voltage (V)	Interference Current Limit for Track Circuit DA Value for Track Circuit Length (A)		
			600m	1200m	1800m
BR939A	20	1.100	4.25	4.16	3.97
BR966 F2	9	0.730	2.96	2.92	2.89
BR966 F9	60	1.900	Not Calculated > 6A		
BI	9	0.531	2.06	2.03	2.01
CAA02	9	0.470	1.78	1.76	1.74
NT2A	9	0.469	1.78	1.75	1.74
D6	9	0.427	1.58	1.57	1.55
CAA04	9	0.290	0.96	0.95	0.95

Table 20 – Four Track Failed Infrastructure (Jointed Rail, Booster Sections, Return Conductor) Calculated Interference Current

6 AXLE TO AXLE VOLTAGES

For a single rail track circuit, occupied by a train, a wrong side failure may be caused by a train on an adjacent track producing an excessive axle to axle voltage, which is transferred via cross bonds (ref. 2.5.7).

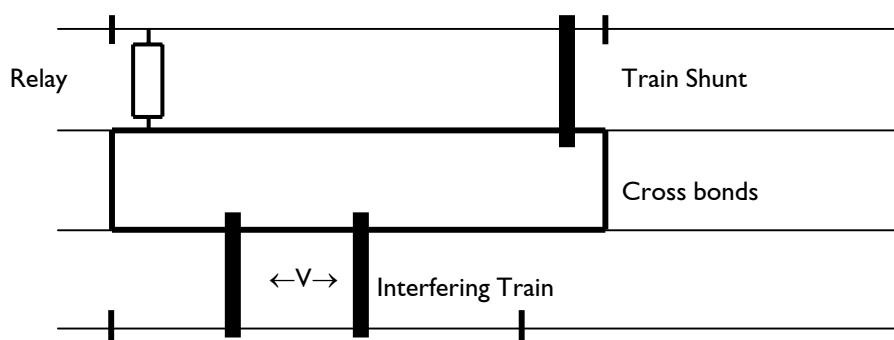


Figure 14 - Typical Axle to Axle Voltage Interference Scenario

The relationship between the voltage developed along the length of the interfering train and that presented across the track relay is dependent upon the rail lengths (and rail resistances) involved and the presence of broken rails or bonding. Furthermore, there are other more complex scenarios involving transposition bonds and physically staggered insulated rail joints.

However, the absolute worst case is for the axle to axle interference voltage to be presented directly across the track relay. Therefore the 'standard' axle to axle voltage limit is simply the drop away voltage of the relay in question, with an appropriate apportionment factor applied to account for interference which may be present from other sources.

7 FREQUENCY RESPONSE

7.1 Relay Response to Varying D.C.

D.c. track relays for use in a.c. electrified areas are immunised by their construction against the effect of a.c. voltages on their input by having a very high impedance to 50Hz traction voltages. However at very low frequencies the relay will follow an a.c. signal, intermittently energising and de-energising. The response to low frequency a.c. depends on the relay type.

Table 2I has been derived empirically and provides multiplication factors which may be applied to the steady state d.c. interference current limit to obtain effective interference limits at frequencies up to 10Hz.

These factors are applicable only to genuine low frequency components, rather than to a low pass filtered asynchronous a.c. currents such as those caused by transformer inrush, OHLE icing and other forms of poor current collection.

These figures are the results of testing and modelling of relays to determine the current scaling required to just pick-up the track relay, and include worst case figures where several sets of results exist, or where relay types are very close in performance characteristic (refs. 2.3.5, 2.3.6, 2.3.7).

Frequency (Hz)	Relay Type			
	BR939A/ BR966F2	BI	D6	CAA04
0 (d.c.)	1.00	1.00	1.00	1.00
1	1.14	4.46	3.94	3.21
2	1.95	23.40	9.84	8.44
3	3.09	53.49	17.72	16.88
4	4.57	102.51	27.56	33.77
5	6.17	-	36.09	54.65
6	8.68	-	49.22	80.37
7	10.97	-	68.91	112.51
8	14.17	-	91.88	152.70
9	17.14	-	111.56	192.88
10	20.57	-	141.09	241.12

Table 21 - Low Frequency Behaviour of d.c. Relays.

Where the d.c. relay type is not known, the characteristic for the BR939A/BR966F2 (refs. 2.5.3, 2.5.4) should be used, as this is considered to give the widest bandwidth. This is also presented in graphical form in Appendix F.

When applying the factors in the above table, it should be recognised that exceedance of the adjusted interference current limit indicates that there is a risk of intermittent false operation of the track relay. The significance of such intermittent operation will depend upon the use of the track relay within the signalling system. Where, in accordance with current best practice, slow to pick repeat relays or TPRs (to BR933A, ref. 2.5.5), are fitted, or equivalent interlocking delay applied (RT/E/PS/11755, ref. 2.2.2), the track relay would need to be energised for sufficient time for a change of track circuit state to be registered, before there was any possibility of interference with the signalling system.

Therefore where slow to pick TPRs are fitted, or equivalent interlocking delay applied, the scaling factors which are applicable become much greater, effectively immunising the track circuit against low frequency interference. In this case, any very slowly varying d.c. content of an interference signal impressed across a d.c. track relay, needs to be considered as a series of transients (see Section 8).

7.2 Frequency Response for Rolling Stock Measurements

When measuring d.c. and low frequency interference levels on rolling stock, it is necessary to filter out the effective 'd.c.' content, which at most is expected to be of the order of a few amps (and usually much lower), against a background of hundreds of amps of a.c. traction current.

An exercise to determine how much of the 50Hz a.c. is rejected by the d.c. track relay has been undertaken and is reported in ref. (2.3.10). This shows that the effective d.c. content on the application of a 50Hz signal is extremely low, in the range of -54 to -67 dB of the 50Hz current.

Therefore to ensure sufficient rejection of the main traction current, and for the purposes of quantifying the d.c. content of the a.c. traction waveform, the attenuation of the measurement channel at 50Hz should be in the range of -54 to -67 dB, ref. (2.5.7).

8 TRANSIENTS

8.1 Approach

Transient d.c. components in the train current are typically produced by the train's transformer as a result of discontinuities in the train current. These tend to be in the form of asymmetric a.c. currents. Therefore the effective d.c. component to be considered will depend upon the filtering applied in the measurement, which should be matched to the track relay types relevant to the route.

The most significant transients are due to transformer inrush and whilst it may be possible to demonstrate compatibility from first principles for limited routes (with no low voltage relays and jointed rail track), the principal method of demonstrating compatibility will be by comparisons with existing rolling stock. In order for this comparative approach to be valid, it should be established through analysis and testing that all other transients are less severe and less frequent than transformer inrush.

8.2 Traction Transformer Characterisation

The worst case transients generally encountered by d.c. track circuits are those associated with traction transformer inrush at energisation from the 25kV supply. Compatibility will be demonstrated by showing that the inrush characteristic of the traction transformer (as determined from measurable/calculable parameters) is within acceptable limits.

Worst case transformer inrush occurs when a number of factors come together simultaneously and which must be included in any calculation or testing to characterise the traction transformer performance.

These key factors are:

- maximum supply voltage (29kV) (for maximum flux density);
- peak residual magnetic flux;
- energisation at voltage zero giving peak flux in opposite direction to residual flux;
- distance to feeder station;
- source impedance of feeder station;
- train impedance.

For tests to represent a reasonable worst case, they should be conducted with the train close to a strong feeder station to minimise the effective source impedance. Provided that the OHLE supply voltage is recorded during the tests, then the expected inrush current for different supply voltages can be extrapolated from test results.

The time constant of transformer in-rush is determined, to an extent, by the L/R ratio of the power supply transformer primary winding; the greater the resistance the shorter the time constant. Consequently the shortest decay should occur at the farthest distance from the feeder station, where the supply resistance is added to the primary winding resistance. Hence the worst case for transformer inrush, in terms of both amplitude and duration, occurs with the train adjacent to the feeder station.

Whilst the initial peak value of the inrush will vary depending upon the instantaneous supply voltage and the residual flux in the train's main transformer core at the instant the inrush occurs, the decay shape of the inrush waveform depends upon the characteristics of the transformer. To ensure that the maximum inrush value can be determined from statistical analysis of the results, then at least 50 inrushes should be recorded.

8.3 Permissible Levels for Transformer Inrush

Transformer inrush is a normal event that is common to all electric traction operating on the a.c. railway. The peak current and d.c. content of an inrush event can be determined from the design of the main transformer and validated by testing. Whilst it is possible to characterise a train in respect of inrush events, no specific acceptance criteria in respect of transformer inrushes can be established. What is known is that existing electric traction appears to operate with acceptable levels of safety over d.c. track circuits.

This code of practice therefore requires that a risk based approach be taken to demonstrate compatibility with d.c. track circuits in respect of transformer inrush events. This should be based on a comparison of the inrush characteristics of new a.c. electric traction against the inrush characteristics of electric traction which already operates on the a.c. railway.

The main features of the design that serve to limit the magnitude of the peak inrush current and its time constant should be discussed. Where there is the potential for simultaneous closure of two or more VCBs then this should be prevented by staggered VCB closure. A delay between successive VCB closure events sufficient to ensure that the resulting inrush events occur in isolation must be incorporated so that the effects of the preceding inrush event will have dissipated to a significantly negligible level.

9 OHLE ICING

Ice on the catenary wire affects all a.c. electric traction and can result in a d.c. offset being produced as a result of the discontinuity that the ice presents between the OHLE and the current collector (Refs. 2.3.8, 2.3.9).

The net effect is that levels of d.c. can be produced that will exceed the maximum permissible steady state levels defined in section 5 of this standard. Since ice on the OHLE is a normal event during periods of cold weather this is a simple matter of fact and will occur.

As far as ensuring compatibility with d.c. track circuits under these conditions it is necessary to ensure that the primary transformer d.c. resistance of the train is comparable to those already using the route. This will ensure that the risk will not be increased by introduction of the new or modified rolling stock.

In addition to this, protection provided by current monitoring devices should be considered and included if this is necessary to meet the ALARP principle.

10 APPORTIONMENT FACTORS

As discussed in section 5, the calculations and the resultant levels relate to the current required to produce a voltage across the relay equal to its drop away voltage. No apportionment factor has been included in any of the calculations.

To account for background levels and provide a suitable margin of safety, the levels defined in section 5 must be divided by an appropriate apportionment factor:

- a factor of 3 for intact infrastructure conditions;
- a factor of 2 for failed infrastructure conditions.

For the single track failed infrastructure condition other arguments may be employed in lieu of apportioning further a level which is already very low:

- The worst case scenario modelled assumes that return conductors are only connected to the return rail at a mid-point connection (MPC), and considers a railbreak between the train and the MPC within the same track circuit. This is an order of magnitude less likely than the worst case scenario considered for double track layouts.
- The worst case scenario modelled assumes an effective run-round path of over 6km, with a series resistance of over 1Ω . For even modest a.c. train currents, this is likely to result in a 50Hz voltage across the relay terminals that exceeds the specified immunity level (75V – refs. 2.5.3, 2.5.4) and will reduce the d.c. sensitivity level.
- On single track sections, the effect of other train interference sources can be expected to be less than in the double track faulty infrastructure case.

In practice, therefore, the applicable limit for single track lines with faulty infrastructure is no lower than that for double track lines (ref. 2.3.3).

For axle to axle voltages, an apportionment factor of 2 is appropriate, as the worst case scenario, where the whole of the axle to axle voltage is presented across a track relay, will only occur under infrastructure failure conditions.

APPENDIX A: PARAMETERS

Defined in Refs. 2.5.1 & 2.5.2, and summarised in section 4.

APPENDIX B: ASSUMPTIONS

Defined in Refs. 2.5.1 & 2.5.2, and summarised in section 4.

APPENDIX C: HAZARDS

The following table lists the hazards identified that are relevant to the a.c. immune d.c. track circuit equipment.

Hazard	Close out
Interference current in the traction return.	Covered by the assessment as presented in refs. 2.5.1 & 2.5.2 and summarised in this document.
Axle to axle or longitudinal voltages.	Covered as presented in this document.
Transient events.	Covered as presented in this document.
Interference from external events (e.g. discontinuities between current collector and OHL).	Covered as presented in this document.

APPENDIX D: MITIGATING FACTORS

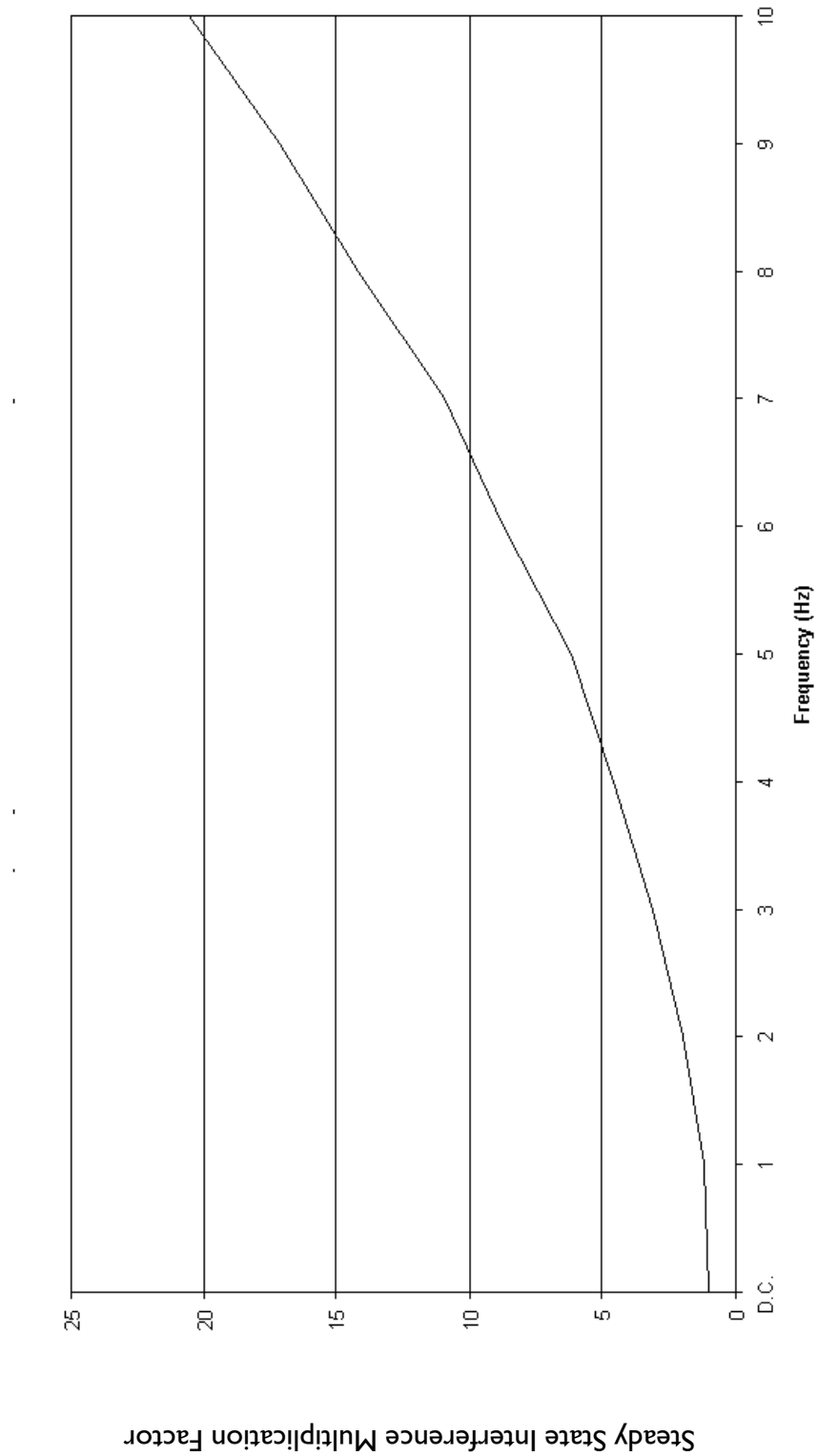
As defined in the main document.

APPENDIX E: SOURCES OF TECHNICAL WORK

The technical work contained in this document is taken from a number of sources:

- The Class 333 Safety Case, produced by Siemens. ISA was WS Atkins Rail.
- The Class 334 Safety Case, produced by AEA Technology Rail and Alstom Transport. ISA was WS Atkins Rail.
- The Class 357 Safety Case, produced by Adtranz. ISA was AEA Technology Rail.
- The Class 373/2 Safety Case, produced by AEA Technology Rail. ISA was Mott MacDonald.
- The Class 390 'M' Stage Safety Case, produced by Alstom Transport. ISA was WS Atkins Rail.
- Modelling conducted by Network Rail, reviewed by WS Atkins Rail

APPENDIX F: LOW FREQUENCY CHARACTERISTIC OF
BR939A/BR966F2 RELAYS



NETWORK RAIL

BRIEFING NOTE

RT/E/C/50004

Issue I

**Methodology for the demonstration of
Electrical Compatibility With DC
(AC Immune) Track Circuits**

Issue Date: April 2003

Background / Synopsis

This standard is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Network Rail's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a one of a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
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